Learning rate experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Learning rate experiments: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Run a **controlled comparison** of **learning rate**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Batching and epochs

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an example a batch an update and an epoch

Reference example, not the only acceptable wording: Teacher explanation: Batch lengths 4, 4, 2; three updates per epoch, nine total updates and 30 example visits.

Updating parameters

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a **learning rate** to an ordered gradient

Reference example, not the only acceptable wording: Teacher explanation: New a = 2 - 0.2 times -3 = 2.6.

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate per-example squared error and mean squared error

Reference example, not the only acceptable wording: Teacher explanation: Residuals are -1 and 2, squares 1 and 4, MSE 2.5.

Use two 45-minute sessions and connect signed distance to squared loss before introducing the factor formula. This is a fully explained quadratic experiment with a frozen hidden representation.

Avoid general prescriptions about rate values. Students should distinguish an oscillating parameter from increasing loss.

Outcomes

Run a **controlled comparison** of **learning rate**s

Explain slow progress **oscillation** and increasing loss in a toy model

Report the limits of a **training** experiment

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

An epoch visits the data once, but each batch causes an update. More batches mean more updates under the stated rule.

End goal for the pictorial case

By the end, I can explain **learning rate** experiments using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Learning rate experiments using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Learning rate experiments · 1

Use the same restricted network h = ReLU(x), y = vh, with only v trained. Keep the fictional **training** pairs (1, 2) and (2, 4), start v = 0, and use one full-data batch per update.

Compare rates 0.05, 0.2, 0.3, 0.4 and 0.5 for exactly four updates each. Restart at zero for every run.

The **training** objective is ordinary MSE: L(v) = 2.5 times (v - 2) squared. Its gradient is 5 times (v - 2). A step is new v = v - rate times 5 times (v - 2).

This small quadratic lets us explain each result rather than treating the rate as a magic setting.

The same rate need not behave this way for another model, **input** scale, loss convention or batch rule. Rate 0.2 is an exact one-step solution here because of this particular gradient formula.

Explain Learning rate experiments · 2

Write d = v - 2, the signed distance from this objective minimum. Substituting the update gives new d = (1 - 5 times rate) times old d. At rate 0.05 the factor is 0.75: distance shrinks without changing sign.

At rate 0.3 it is -0.5: distance changes sign and halves. Both can reduce loss.

Explain Learning rate experiments · 3

The helper assumes a finite rate and nonnegative integer update count. Its five logged states include the initial value.

At 0.3 the parameter sequence is [0, 3, 1.5, 2.25, 1.875] and loss is [10, 2.5, 0.625, 0.15625, 0.0390625]. Crossing the minimum does not itself mean **training** failed.

Term	Meaning
Learning rate	The multiplier controlling the size of a gradient update

Term	Meaning
Controlled comparison	An experiment changing a named factor while holding other stated conditions fixed
Oscillation	Repeated movement from one side of a value to the other
Divergence	Movement away from a target solution with growing error in this experiment

The update factor explains three behaviours

RATE AND FACTOR	SIGNED DISTANCE BEHAVIOUR	TRAINING LOSS BEHAVIOUR
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

Explain Learning rate experiments · 4

Guided: multiplier -1; v becomes 4 then 0, with MSE 10 at both states. Parameter change does not guarantee decreasing loss.

Exit: for the fixed two-example objective, start and four-update budget, rate 0.2 reached zero **training** MSE after one update. This does not demonstrate an optimal rate or generalisation for other networks or data.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The update factor explains three behaviours

RATE AND FACTOR	SIGNED DISTANCE BEHAVIOUR	TRAINING LOSS BEHAVIOUR
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Learning rate experiments

Compare **learning rate**s with identical starting loss and update counts.

The facts we will use

Supplied facts

Loss $J(a)=(a-4)^2$; gradient $2(a-4)$; start a = = 0. Compare two updates at rates0.25 and 0.75.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/14

Learning rate experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Rate0.25

First a = 2, loss 4; second a = 3, loss 1.



2. Rate0.75

First a = 6, loss 4; second a = 3, loss 1.



3. Inspect paths

Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

CHANGE ONE THING

Add a rate 1 run from a = 0.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Rate0.25	First a = 2, loss 4; second a = 3, loss 1.
2. Rate0.75	First a = 6, loss 4; second a = 3, loss 1.
3. Inspect paths	Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

What this example does not establish

Keep the stated limit

These quadratic paths explain step size. Real nonconvex networks can have more

complicated rate behaviour.

Change one thing in the pictured case

Change one thing

Change: Add a rate 1 run from a = 0.

Predicted result

Prediction: Its path is 8 then 0, with loss 16 at both steps.

Why the result follows

Reason: The update alternates across4 without reducing the initial loss.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

TEACHER GUIDE · 3 OF 9

Demonstrate and guide practice

Demonstrate Learning rate experiments, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve epoch and update counts; agree on one full-data batch and four updates per run.
7 to 18 minutes	Calculate the first row of each rate run and keep all resets visible.
18 to 32 minutes	Derive new d from the update rule using substitution; complete W1.
32 to 45 minutes	Run or trace Python and compare rate 0.3 with W2, marking sign and magnitude separately.
Session 2 start to 8 minutes	Complete the guided rate 0.4 example and explain why changing values can leave loss unchanged.
8 to 20 minutes	Use W3 to connect the loss scale with the gradient and an equivalent rate.
20 to 35 minutes	Collect the three independent W4 traces with a fresh reset before each.
35 to 45 minutes	Use W5 and the exit response, then administer the fresh retry and record the experimental limits.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a number line with the minimum marked and a table for old value, gradient, new value and loss.

Extension: derive 0 < rate < 1 for strict loss decrease on J(a) = (a - 1) squared away from the minimum, and compare with the lesson bound 0.4.

Reduce help deliberately

Completion practice

Completion task: At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

Reasoning to listen for

Teacher check: Teacher explanation: Next gradient 5 gives v = 1.5 and loss 0.625. No. Signed distance changes side but shrinks by half each update; loss decreases.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Learning rate** experiments. Proposed explanation: “Crossing the minimum means **divergence**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Compare distance magnitudes or losses; alternating sides can converge.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Use the same objective, start a = = 0 and two-update budget; report full paths, not only the preferred run’s endpoint. Changed-case check: Its path is 8 then 0, with loss 16 at both steps.

The update alternates across4 without reducing the initial loss.

Changed-case reasoning: Its path is 8 then 0, with loss 16 at both steps. The update alternates across4 without reducing the initial loss.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Learning rate experiments investigation and use the first unsupported step to choose a teaching response.

Crossing the minimum means **divergence**: Compare distance magnitudes or losses; alternating sides can converge.

A larger rate is always faster: The tested 0.5 rate increases this objective; 0.4 keeps it unchanged.

The best rate is independent of loss scaling: Scaling the loss scales its gradient and changes the step at a fixed rate.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Learning rate experiments questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: multiplier -1; v becomes 4 then 0, with MSE 10 at both states. Parameter change does not guarantee decreasing loss.

Exit: for the fixed two-example objective, start and four-update budget, rate 0.2 reached zero **training** MSE after one update. This does not demonstrate an optimal rate or generalisation for other networks or data.

W1 Trace slow progress

Workbook question · W1

For rate 0.05, start v = 0 and compute two updates and their **training** MSE values using the lesson objective.

Teacher reasoning · W1

Teacher explanation: First v = 0.5 with MSE 5.625. Second gradient -7.5 gives v = 0.875 and MSE 3.1640625. Distance is multiplied by 0.75 each step.

W2 Explain a crossing

Workbook question · W2

At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

Teacher reasoning · W2

Teacher explanation: Next gradient 5 gives v = 1.5 and loss 0.625. No. Signed distance changes side but shrinks by half each update; loss decreases.

W3 Preserve an update after rescaling loss

Workbook question · W3

Replace MSE by the sum of squared errors on the two **training** pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

Teacher reasoning · W3

Teacher explanation: The sum gradient is twice the mean gradient, so rate 0.15 reproduces the original steps. Rate 0.3 would double each step from the same state.

W4 Compare a new objective independently

Workbook question · W4

Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

Teacher reasoning · W4

Teacher explanation: At 0.25: a = 0.5 then 0.75; losses 0.25 then 0.0625. At 0.5: a = 1 then 1; losses 0 then 0. At 1: a = 2 then 0; losses 1 then 1. Rate 0.5 attains the lowest objective in this comparison.

The useful rate differs from the lesson model.

W5 Find an unfair comparison

Workbook question · W5

Run A starts v = 0 with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final **training** losses. Can the difference be attributed only to rate?

Propose a repair.

Teacher reasoning · W5

Teacher explanation: No. Initial state and batch rule also changed. Restart every run at the same v, hold data, loss, batch rule and update budget fixed, and vary only rate.

Report the traces and separate any later experiment with a different factor.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 must include all six new states and losses with no carry-over between runs. Fresh retry: J(c) = 2(c - 3) squared, gradient 4(c - 3), start c = 1. At rate 0.125, c becomes 2 then 2.5 with losses 2 then 0.5.

At rate 0.25, c becomes 3 then 3 with losses 0 then 0. Ask for a conclusion limited to these runs.

Assess independent transfer

Independent question · withhold the key

Selected case: Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run.

Identify the result supported after two updates.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: At 0.25: a = 0.5 then 0.75; losses 0.25 then 0.0625. At 0.5: a = 1 then 1; losses 0 then 0. At 1: a = 2 then 0; losses 1 then 1. Rate 0.5 attains the lowest objective in this comparison.

The useful rate differs from the lesson model.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Learning rate experiments to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **learning rate** experiments **evidence** sheet to Image-pattern Network Investigation. Show how you can run a **controlled comparison** of **learning rate**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a number line with the minimum marked and a table for old value, gradient, new value and loss.

Extension: derive 0 < rate < 1 for strict loss decrease on J(a) = (a - 1) squared away from the minimum, and compare with the lesson bound 0.4.

Week 15 separates **training** and validation curves.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/15, ai-5/17, ai-5/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Learning rate experiments lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Run a **controlled comparison** of **learning rate**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

Supported practice answer

Reasoned practice answer: Teacher explanation: Next gradient 5 gives v = 1.5 and loss 0.625. No. Signed distance changes side but shrinks by half each update; loss decreases.

Check every learner before discussion: Use the worked case for **Learning rate** experiments. Proposed explanation: “Crossing the minimum means **divergence**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Compare distance magnitudes or losses; alternating sides can converge.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Add a rate 1 run from a = 0. Require a **prediction** and a reason before comparing. Expected result: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across4 without reducing the initial loss.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

TEACHER GUIDE · 9 OF 9

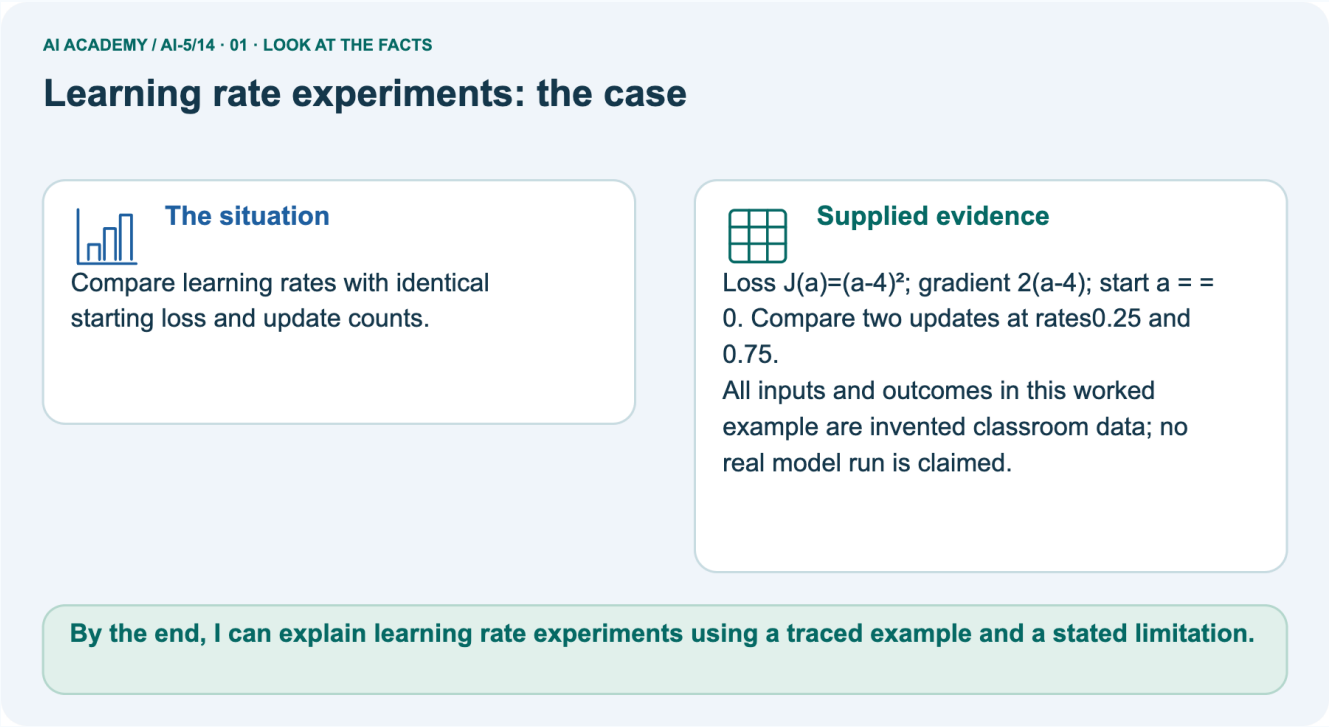
Visual concept journey

> Show one Learning rate experiments graphic at a time, request a prediction and discuss its

> trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Learning rate experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare **learning rate**s with identical starting loss and update counts. Supplied facts: Loss J(a)=(a-4)²; gradient 2(a-4); start a = = 0. Compare two updates at rates0.25 and 0.75.

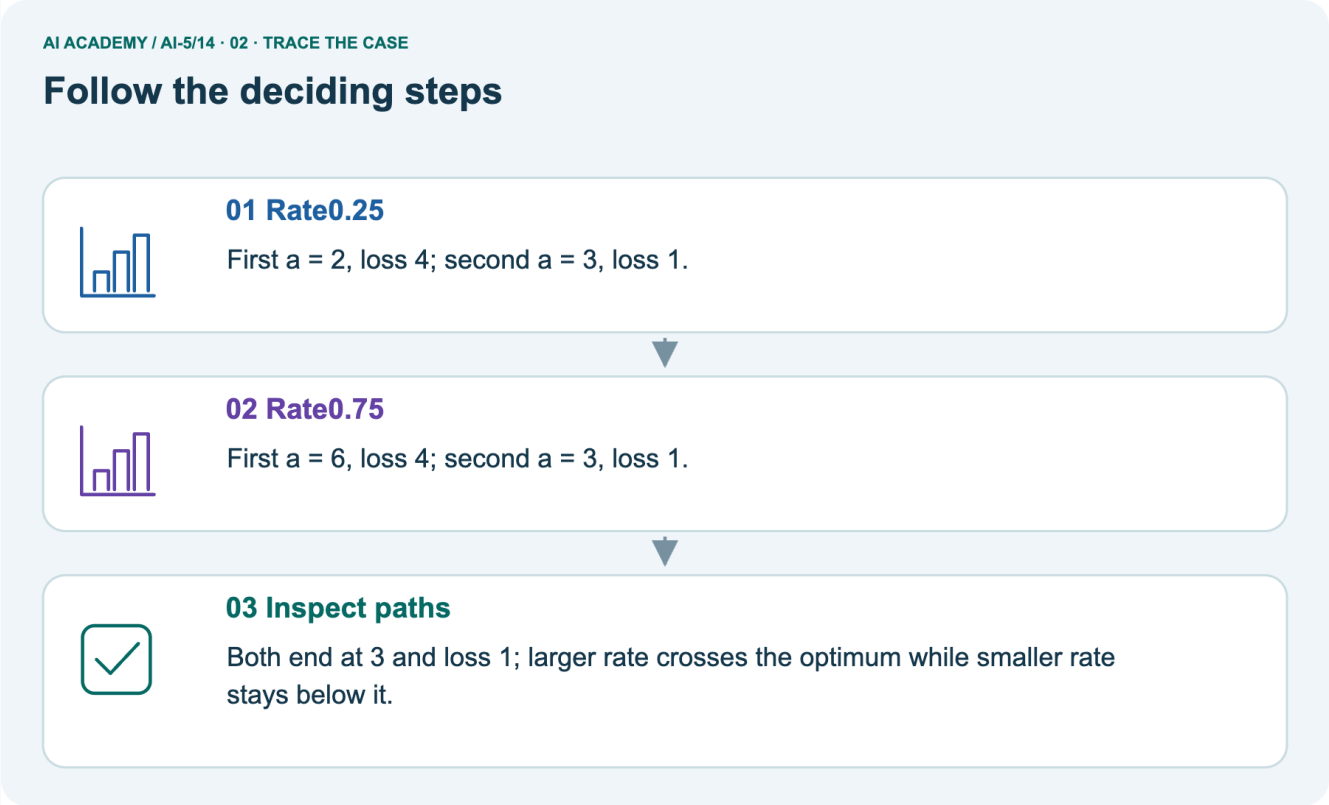
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

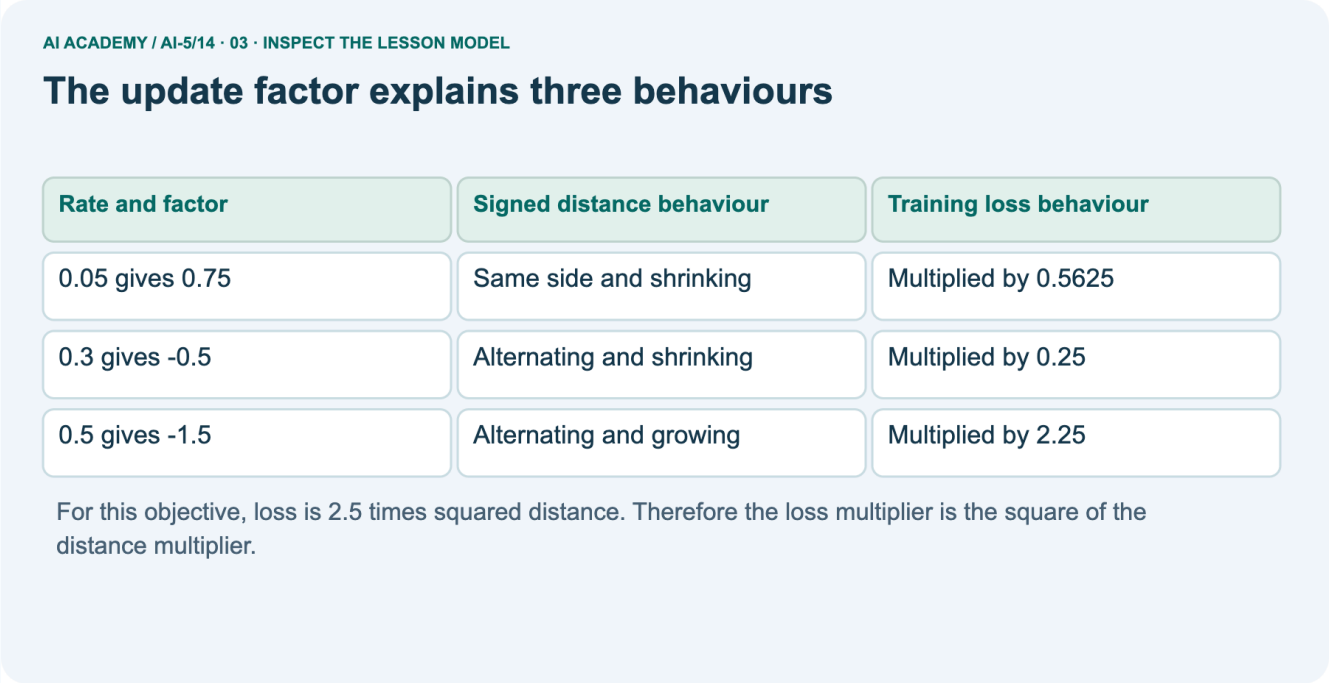
Rate0.25: First a = 2, loss 4; second a = 3, loss 1. → Rate0.75: First a = 6, loss 4; second a = 3, loss 1. → Inspect paths: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. The update factor explains three behaviours



Follow the labelled picture with the matching explanation. Open the vector diagram

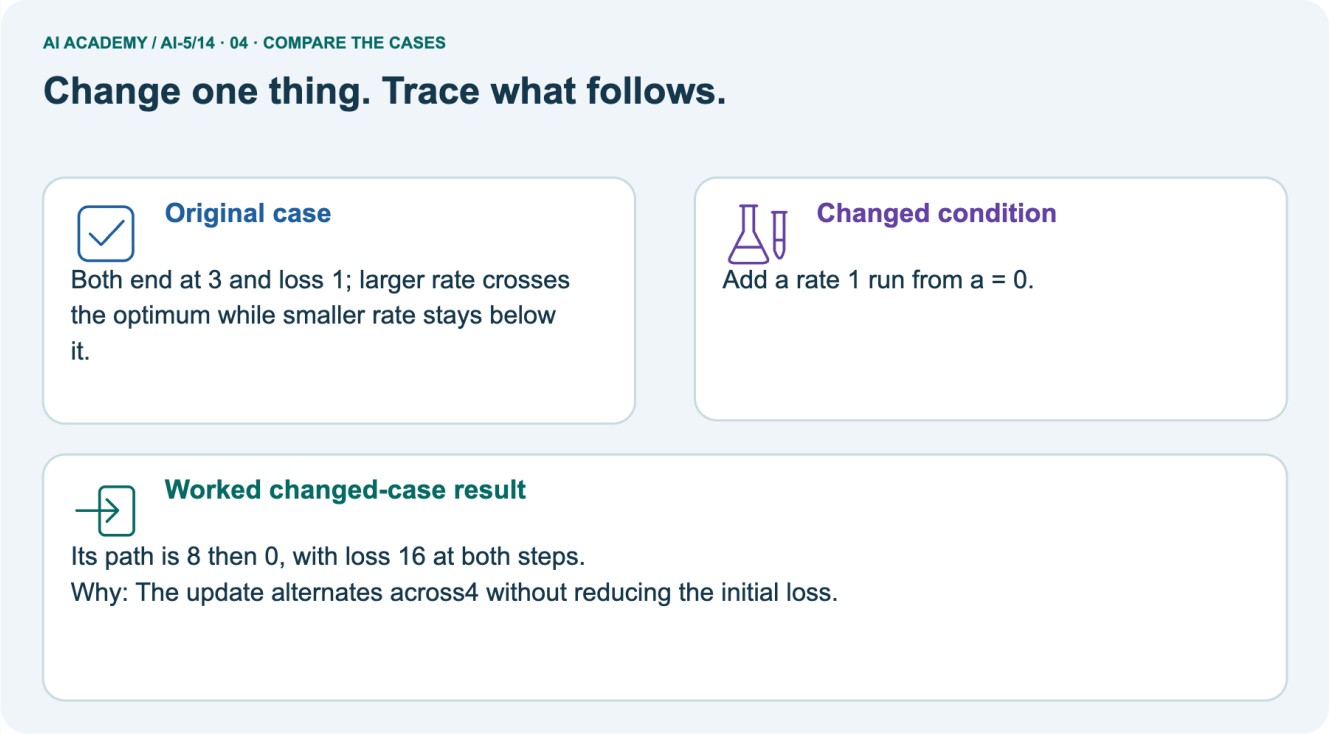
The update factor explains three behaviours. 0.05 gives 0.75 | Same side and shrinking | Multiplied by 0.5625. 0.3 gives -0.5 | Alternating and shrinking | Multiplied by 0.25. 0.5 gives -1.5 | Alternating and growing | Multiplied by 2.25

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it. Changed condition: Add a rate 1 run from a = 0. Expected result: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across4 without reducing the initial loss.

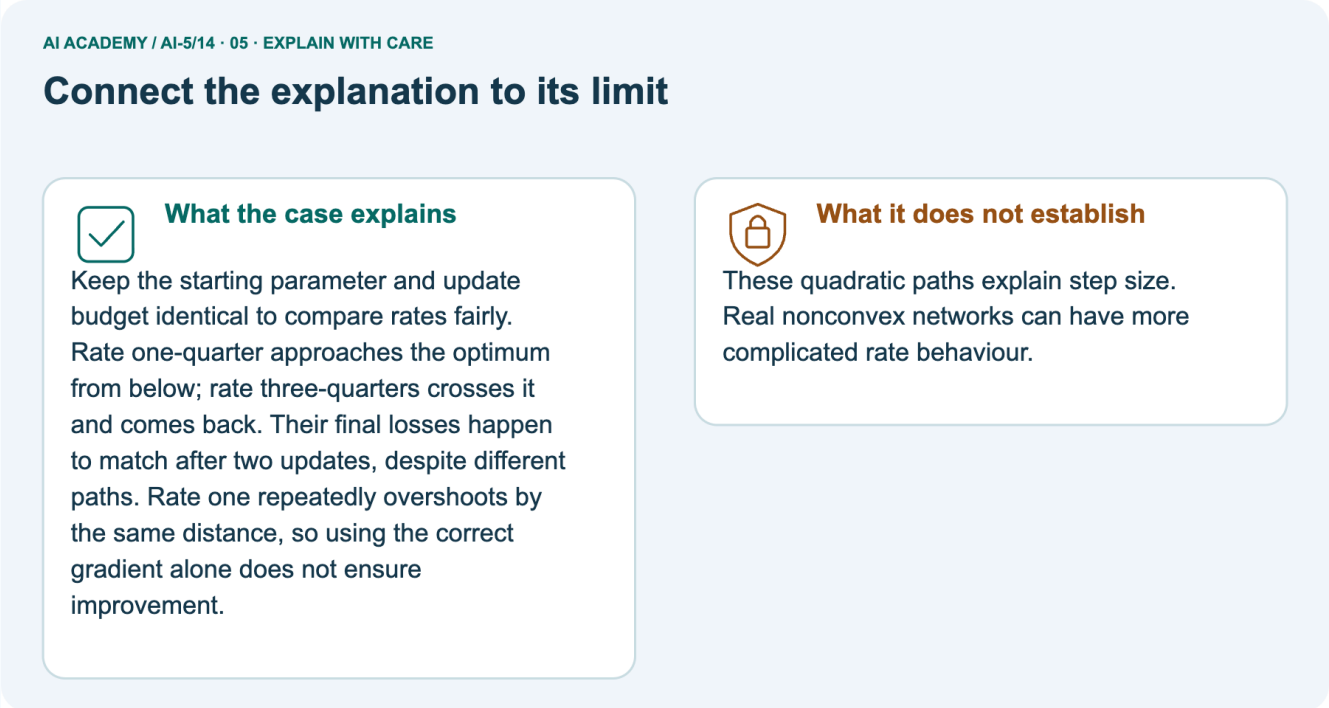
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The update alternates across4 without reducing the initial loss.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Limit: These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

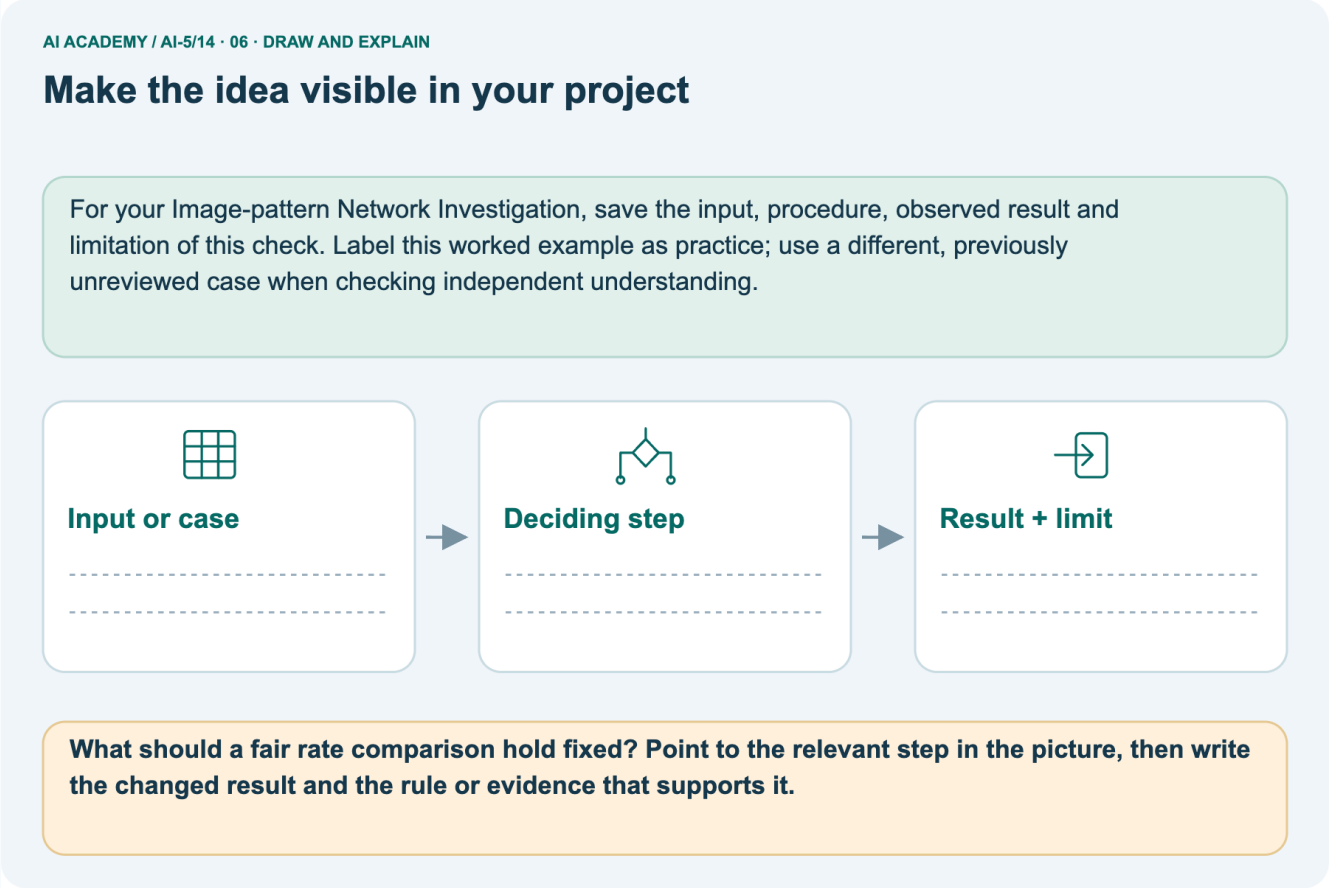
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Use the same objective, start a = = 0 and two-update budget; report full paths, not only the preferred run’s endpoint. Changed-case check: Its path is 8 then 0, with loss 16 at both steps.

The update alternates across4 without reducing the initial loss.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.